

# RETAIL SALES ANALYTICS DASHBOARD

## Project Title

Retail Sales Analytics Dashboard using SQL, Python, and Power BI

### 1. Abstract

This project focuses on analyzing retail sales data to identify business performance, customer behavior, product profitability, and regional sales trends. The project combines SQL for data querying, Python for data analysis and visualization, and Power BI for interactive dashboard development.

The dashboard provides insights into total sales, profit, orders, product performance, customer segments, category-wise sales contribution, and regional distribution. The objective is to help business stakeholders make data-driven decisions and improve overall sales performance.

### 2. Project Objectives

The main objectives of this project are:

- Analyze retail sales performance.
- Identify top-performing products.
- Evaluate category-wise profitability.
- Understand customer segment behavior.
- Compare sales across different regions.
- Monitor overall business performance using KPIs.
- Create an interactive dashboard for decision-making.

### Project Description

#### Retail Sales Analytics Dashboard | SQL, Python, Power BI

- Analyzed 9,994 retail transactions to identify sales, profit, customer, and product performance trends.
- Performed data extraction and business analysis using MySQL.
- Conducted exploratory data analysis and visualization using Python libraries including Pandas, Matplotlib, and Seaborn.
- Developed an interactive Power BI dashboard featuring KPI cards, sales trends, customer segmentation, regional analysis, and product performance metrics.
- Generated actionable business insights to support strategic decision-making and improve profitability.

### 3. Tools and Technologies Used

| Tool  | Purpose                  |
|-------|--------------------------|
| MySQL | Data Querying & Analysis |

| Tool       | Purpose                       |
|------------|-------------------------------|
| Python     | Data Cleaning & Visualization |
| Pandas     | Data Manipulation             |
| NumPy      | Numerical Computations        |
| Matplotlib | Data Visualization            |
| Seaborn    | Statistical Visualization     |
| Power BI   | Dashboard Development         |
| GitHub     | Project Documentation         |

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## 4. Dataset Description

Dataset Used:

### Sample Superstore Dataset

The dataset contains retail sales transactions including:

- Order ID
- Order Date
- Customer ID
- Segment
- Region
- State
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Discount
- Profit

The dataset consists of approximately 9,994 sales transactions.

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## 5. SQL Analysis

SQL was used to perform business analysis and answer key business questions.

Database Used

superstore\_db

Table Used

sample\_superstore

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## SQL Business Questions

### Total Sales

```
SELECT ROUND(SUM(Sales),2) AS Total_Sales  
FROM sample_superstore;
```

Purpose:

To determine total revenue generated by the business.

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### Total Profit

```
SELECT ROUND(SUM(Profit),2) AS Total_Profit  
FROM sample_superstore;
```

Purpose:

To evaluate overall profitability.

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### Sales by Region

```
SELECT Region,  
ROUND(SUM(Sales),2) AS Revenue  
FROM sample_superstore  
GROUP BY Region  
ORDER BY Revenue DESC;
```

Purpose:

To identify top-performing regions.

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### Category-wise Sales

```
SELECT Category,  
ROUND(SUM(Sales),2) AS Revenue  
FROM sample_superstore  
GROUP BY Category  
ORDER BY Revenue DESC;
```

Purpose:

To determine which product category contributes the highest revenue.

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### Top 10 Products by Sales

```
SELECT Product_Name,  
ROUND(SUM(Sales),2) AS Revenue  
FROM sample_superstore  
GROUP BY Product_Name  
ORDER BY Revenue DESC  
LIMIT 10;
```

ORDER BY Revenue DESC

LIMIT 10;

Purpose:

To identify the most profitable products.

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### Customer Segment Analysis

```
SELECT Segment,  
ROUND(SUM(Sales),2) AS Revenue  
FROM sample_superstore  
GROUP BY Segment;
```

Purpose:

To understand customer purchasing behavior.

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### 6. Python Analysis

Python was used for exploratory data analysis and visualization.

Libraries Used

```
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns
```

Key Activities Performed

- Dataset loading
  - Data inspection
  - Missing value analysis
  - Sales analysis
  - Profit analysis
  - Category analysis
  - Regional performance analysis
  - Data visualization
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### 7. Power BI Dashboard Development

The final stage of the project involved developing an interactive Power BI dashboard to visualize key business metrics and provide actionable insights.

The dashboard was designed to help stakeholders quickly monitor sales performance, profitability, customer behavior, and regional trends through interactive charts, filters, and KPI cards.

The dashboard enables users to filter data dynamically by Region, Category, Segment, and Year for deeper analysis.

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### 8. Dashboard Components

### 8.1 KPI Cards

The KPI cards provide a quick overview of the company's performance.

#### Total Sales

Displays the total revenue generated by the business.

**Value:** \$2.30 Million

#### Total Profit

Displays the total profit earned after accounting for costs.

**Value:** \$286.40 Thousand

#### Total Orders

Shows the total number of customer orders processed.

**Value:** 9,994 Orders

#### Total Quantity Sold

Displays the total number of products sold.

**Value:** 38 Thousand Units

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### 8.2 Region Wise Sales Analysis

#### Visualization Used

Bar Chart

#### Purpose

To compare sales performance across different geographical regions.

#### Observation

The West region generated the highest sales revenue, followed by East and Central regions.

#### Business Impact

Helps management identify high-performing regions and areas needing improvement.

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### 8.3 Category Contribution to Sales

#### Visualization Used

Donut Chart

#### Purpose

To analyze the contribution of each product category to total sales.

#### Observation

Technology contributed the highest share of total sales.

#### Business Impact

Helps prioritize inventory planning and marketing investments.

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## 8.4 Profit by Category

### Visualization Used

Column Chart

### Purpose

To compare profitability among different product categories.

### Observation

Technology generated the highest profit, while Furniture contributed comparatively lower profits.

### Business Impact

Supports strategic decisions regarding product portfolio optimization.

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## 8.5 Sales vs Profit Analysis

### Visualization Used

Scatter Chart

### Purpose

To analyze the relationship between sales and profitability.

### Observation

Some categories generated high sales but lower profit margins.

### Business Impact

Helps identify products requiring pricing or cost optimization.

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## 8.6 Monthly Sales Trend

### Visualization Used

Line Chart

### Purpose

To monitor sales performance over time.

### Observation

Sales remained relatively stable with periodic fluctuations.

### Business Impact

Supports demand forecasting and seasonal planning.

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## 8.7 Customer Segment Analysis

### Visualization Used

Pie Chart

### Purpose

To analyze sales contribution from different customer segments.

### **Observation**

Consumer segment generated the highest revenue contribution.

### **Business Impact**

Helps identify the most valuable customer group.

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## **8.8 Top 10 Products by Sales**

### **Visualization Used**

Horizontal Bar Chart

### **Purpose**

To identify the best-selling products.

### **Observation**

A small number of products generated a significant portion of revenue.

### **Business Impact**

Helps optimize inventory management and promotional campaigns.

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## **8.9 State Wise Sales Distribution**

### **Visualization Used**

Map Visualization

### **Purpose**

To visualize sales distribution geographically.

### **Observation**

Sales activity was concentrated in major states and metropolitan areas.

### **Business Impact**

Supports territory management and regional expansion strategies.

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## **9. DAX Measures Used**

The following DAX measures were created in Power BI for KPI calculations.

### **Total Sales**

Total Sales =  
SUM('Sample\_Superstore'[Sales])

### **Total Profit**

Total Profit =  
SUM('Sample\_Superstore'[Profit])

### **Total Orders**

Total Orders =  
DISTINCTCOUNT('Sample\_Superstore'[Order ID])

### **Total Quantity Sold**

Total Quantity Sold =  
SUM('Sample\_Superstore'[Quantity])

### **Average Order Value**

Average Order Value =  
DIVIDE(  
SUM('Sample\_Superstore'[Sales]),  
DISTINCTCOUNT('Sample\_Superstore'[Order ID])  
)

### **Profit Margin %**

Profit Margin % =  
DIVIDE(  
SUM('Sample\_Superstore'[Profit]),  
SUM('Sample\_Superstore'[Sales])  
) \* 100

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## **IN SIMPLE TERMS,**



### **Power BI Dashboard**

The dashboard includes:

#### **KPI Cards**

- Total Sales
- Total Profit
- Total Orders
- Total Quantity Sold

#### **Visualizations**

- Region Wise Sales Analysis
- Category Contribution to Sales
- Profit by Category
- Monthly Sales Trend
- Customer Segment Analysis
- Sales vs Profit Analysis
- Top 10 Products by Sales
- State Wise Sales Distribution

#### **Filters**

- Segment
- Region
- Category
- Year



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## ✓ End-to-End Data Analytics Workflow

- Total Sales = SUM(Sales)
- Total Profit = SUM(Profit)
- Total Orders = DISTINCTCOUNT(Order ID)
- Total Quantity Sold = SUM(Quantity)
- Profit Margin % = DIVIDE(SUM(Profit), SUM(Sales), 0) \* 100
- Average Sales Per Order = DIVIDE(SUM(Sales), DISTINCTCOUNT (Order ID))
- Total Customers = DISTINCTCOUNT (Customer ID)
- Average Sales Per Customer = DIVIDE(SUM(Sales), DISTINCTCOUNT (Customer ID))

### Visualizations Used

- Card Visual
- Clustered Bar Chart
- Clustered Column Chart
- Donut Chart
- Treemap
- Ribbon Chart
- Waterfall Chart
- Scatter Chart
- Map Visual
- Line Chart
- Slicers
- Navigation Buttons

## 10. Key Insights

The following insights were derived from the analysis:

1. The company generated approximately \$2.30 million in sales revenue.
2. Total profit earned was approximately \$286 thousand.
3. The West region contributed the highest sales revenue.
4. Technology was the highest-performing category in terms of both sales and profitability.
5. Consumer customers contributed the largest share of revenue.
6. A limited number of products generated a significant percentage of overall sales.
7. Sales performance varied significantly across states.
8. Certain categories showed lower profit margins despite strong sales volumes.

These insights help management identify opportunities for growth, improve profitability, and optimize resource allocation.

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## 11. Business Recommendations

Based on the analysis, the following recommendations are proposed:

### Increase Focus on High-Performing Regions

The West region generated the highest sales revenue. Similar strategies can be implemented in other regions to improve overall performance.

### Expand Technology Product Portfolio

Technology products contributed the highest sales and profit. Increasing inventory and promotional campaigns for these products may further increase revenue.

### Improve Furniture Category Profitability

Although Furniture generated considerable sales, profit margins were comparatively lower. Pricing and cost optimization strategies should be explored.

### Strengthen Customer Retention Programs

The Consumer segment contributed the highest revenue. Loyalty programs and personalized marketing campaigns can improve customer retention and repeat purchases.

### Optimize Product Inventory

The Top 10 products contributed significantly to sales revenue. Inventory management should prioritize these products to avoid stock shortages.

### Regional Marketing Strategy

State-level sales analysis can be used to identify underperforming areas and implement targeted marketing campaigns.

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## 12. Conclusion

This project successfully analyzed retail sales data using SQL, Python, and Power BI.

SQL was used for data extraction and business query analysis. Python was utilized for data exploration and visualization, while Power BI was used to create an interactive dashboard for business intelligence reporting.

The dashboard provides a comprehensive overview of sales performance, profitability, customer behavior, product performance, and regional trends. The insights generated can support strategic decision-making and help improve business performance.

The project demonstrates practical skills in data analysis, data visualization, dashboard development, and business intelligence reporting.

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## 13. Future Scope

The project can be enhanced further by implementing:

### Predictive Sales Forecasting

Machine Learning models can be used to predict future sales trends.

### Customer Segmentation

Advanced customer segmentation techniques can be applied for personalized marketing.

### Inventory Optimization

Demand forecasting can improve inventory management and reduce stock shortages.

### **Real-Time Dashboard**

Integration with live databases can enable real-time sales monitoring.

### **AI-Based Business Recommendations**

Artificial Intelligence can be used to generate automated recommendations for business growth.

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## **14. Project Outcome**

The project achieved the following outcomes:

- ✓ Developed a complete Retail Sales Analytics Solution
  - ✓ Performed SQL-based business analysis
  - ✓ Conducted exploratory data analysis using Python
  - ✓ Created an interactive Power BI dashboard
  - ✓ Identified key sales and profitability trends
  - ✓ Generated actionable business insights
  - ✓ Improved understanding of business intelligence concepts
  - ✓ Demonstrated end-to-end Data Analytics workflow
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## **15. Skills Demonstrated**

### **Technical Skills**

- SQL
- MySQL
- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI
- Data Cleaning
- Data Transformation
- DAX
- Data Visualization

### **Business Skills**

- Sales Analysis
- Profitability Analysis

- Customer Analytics
- Product Performance Analysis
- Dashboard Development
- Business Intelligence Reporting